NATHANIEL KONG

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PROFESSIONAL SUMMARY

Final-year Computer Science student at RMIT University building full-stack AI applications with React, Python and PostgreSQL. Deployed Analyq, a stock research platform combining LLM APIs, time-series machine learning and automated testing; also delivered two client websites from brief to launch.

PROJECTS

Analyq | Full-Stack AI Stock Research Platform
analyq.vercel.app | github.com/nathanielkong/Analyq
React, TypeScript, Python, FastAPI, PostgreSQL, scikit-learn, Docker, GitHub Actions
- Built and deployed a stock research platform that turns natural-language questions into charts, comparisons and answers grounded in calculated market metrics and news sentiment, integrating Alpaca, Alpha Vantage and Gemini APIs.
- Implemented Google and password authentication with persistent chat history; reused daily research across conversations through PostgreSQL caching and locking, with automated tests confirming report reuse and duplicate-generation prevention.
- Developed logistic regression models for 1-, 3- and 5-session price direction using momentum, volatility and volume features; evaluated through gap-aware walk-forward validation, Brier scores and naive baselines to flag results without demonstrated predictive advantage.
- Shipped with CI/CD: GitHub Actions runs 203 backend tests and container smoke checks before publishing commit-tagged Docker images to GitHub Container Registry; deployed on Vercel (frontend), Render (API) and Neon (PostgreSQL).

ChainGuard - AI Blockchain Investigation Assistant | Final Year Team Project
React | Python / FastAPI | Pydantic | LiteLLM | Blockchain APIs | Docker Compose
- Implemented a chat and CLI investigation workflow with 3 selectable LLM providers (OpenAI, Gemini and Claude), translating user requests into blockchain data retrieval and plain-language explanations.
- Normalised 4 evidence categories - contracts, transactions, token holdings and liquidity - into Pydantic-validated context with fetcher provenance; validated structured assessments and preserved insufficient-evidence outcomes.
- Integrated a pluggable external detector through a typed HTTP API; separated the frontend, backend and detector into 3 Docker Compose services so detection algorithms can be replaced independently.

MatchLens - Explainable Football Match Intelligence | Microsoft Developer Hackathon Project
github.com/nathanielkong/MatchLens
React | TypeScript | Node.js | Express | PostgreSQL | Microsoft Foundry | Azure | Docker
- Built an interactive football analysis platform with pitch replay, match statistics and probability charts; generated 1,516 synthetic matches across 8 fictional teams with separate training, validation and evaluation splits.
- Implemented evidence-backed summaries, conversational Q&A and presenter recaps through Microsoft Foundry, with replay-cutoff filtering, structured-output validation and a separate AI review before publication.
- Developed a Poisson result-probability model that achieved a 1.5% lower Brier score than a static Poisson baseline across 300 held-out synthetic matches evaluated at 7 match-time cutoffs.

EXPERIENCE

Full-Stack Developer (Contract) Jan 2026 - Apr 2026
1Oak Marketing and Agency Company | Malaysia
Client website: The House of Vows - thehouseofvows.netlify.app
- Delivered 2 client websites, including The House of Vows with Next.js, React and TypeScript; translated client feedback into responsive pages and enquiry pathways, then deployed and maintained sites on Netlify.
- Implemented metadata, JSON-LD and an XML sitemap; Lighthouse checks recorded 91 SEO, 91 accessibility and 100 best practices.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C++
Web & Data: React, Next.js, FastAPI, Spring Boot, REST APIs, PostgreSQL, SQLAlchemy, HTML/CSS
AI Tools: scikit-learn, LiteLLM, Gemini/OpenAI APIs, VADER | **Methods:** feature engineering, logistic regression, time-series validation
Testing & Delivery: pytest, JUnit, Git, GitHub Actions, Docker, Docker Compose, Vercel, Render, Neon

EDUCATION

RMIT University | Melbourne, Australia 2023 - 2026
Bachelor of Computer Science
Relevant Coursework: Full Stack Development, Machine Learning, Practical Data Science, Advanced Programming for Data Science, Algorithms & Analysis, C++ Bootcamp

CERTIFICATIONS

AWS Certified Cloud Practitioner | Amazon Web Services | **7 October 2026**
Verify credential | Validation number: cdfbffe42cab47e9b55001370eed2502